SpaceX Falcon 9 Landing Prediction
Capstone Project

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- Goal: Predict if Falcon 9 first stage will land successfully
- 90 Falcon 9 launches analyzed (2010-2020)
- Best model: Logistic Regression with 87% accuracy
- Key finding: Launch site and payload mass are strongest predictors
- Business impact: Estimated $15M savings per reused rocket

Problem:

• SpaceX launch costs: $62M (vs $165M for competitors)
• Savings come from reusing first stage
• Predicting landing success helps estimate costs

Objectives:

1. Collect and clean launch data
2. Perform EDA and create visualizations
3. Build classification models
4. Create interactive dashboard

Data Source: SpaceX API

- Endpoint: https://api.spacexdata.com/v4/launches/past
- Method: Python requests library
- Data format: JSON

Code Example:

```
response = requests.get(spacex_url)
data = pd.json_normalize(response.json())
```

Collected: 107 launches, 42 features

Cleaning Steps:
1. Filtered to single-core launches (no Falcon Heavy)
2. Extracted nested data from:
   - cores (landing outcomes, reuse count)
   - payloads (mass, orbit)
   - rockets (booster version)
   - launchpads (location coordinates)
3. Removed Falcon 1 launches
4. Handled missing PayloadMass (filled with mean: 2,847 kg)

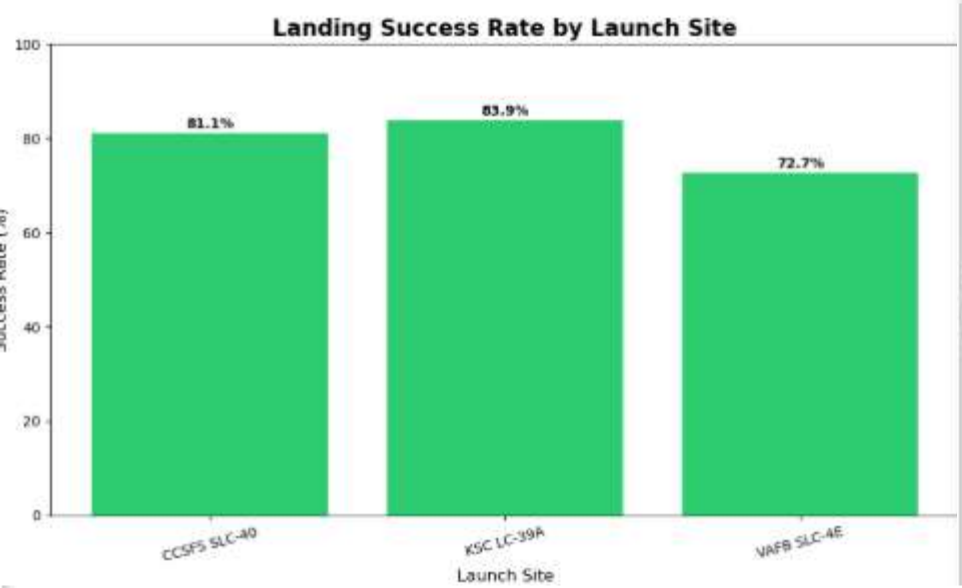
Final dataset: 90 Falcon 9 launches

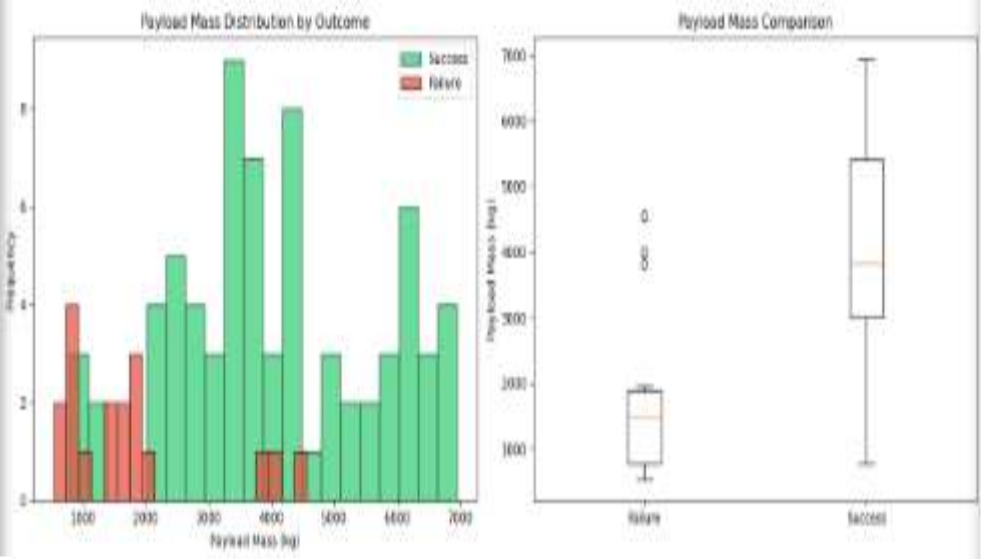
Tools Used:
• Pandas for data manipulation
• Matplotlib & Seaborn for visualizations
• SQL for database queries

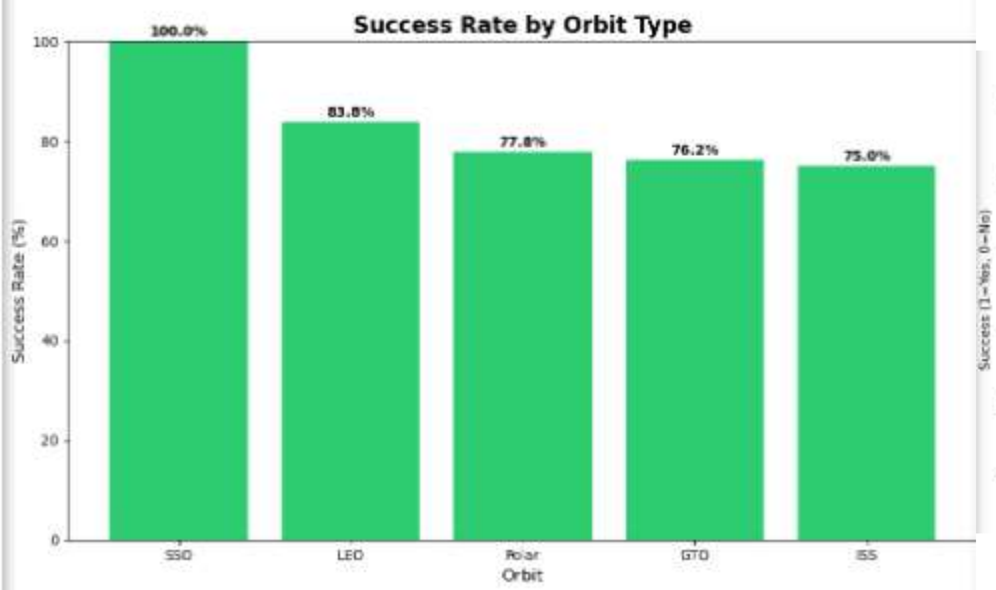
Analysis Techniques:
• Correlation matrices
• Distribution plots
• Success rate by category
• Time series analysis

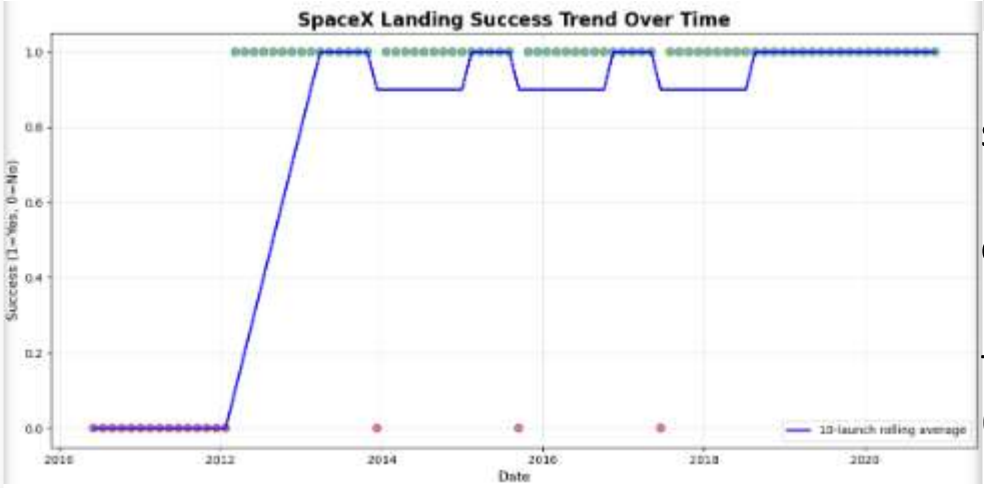
Models Tested:
1. Logistic Regression
2. Support Vector Machine (SVM)
3. Decision Tree
4. K-Nearest Neighbors (KNN)

Process:
• Train/Test split: 70/30
• Features: FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, ReusedCount
• Evaluation: Accuracy, Precision, Recall, F1-Score









Key Insights from Visualizations:

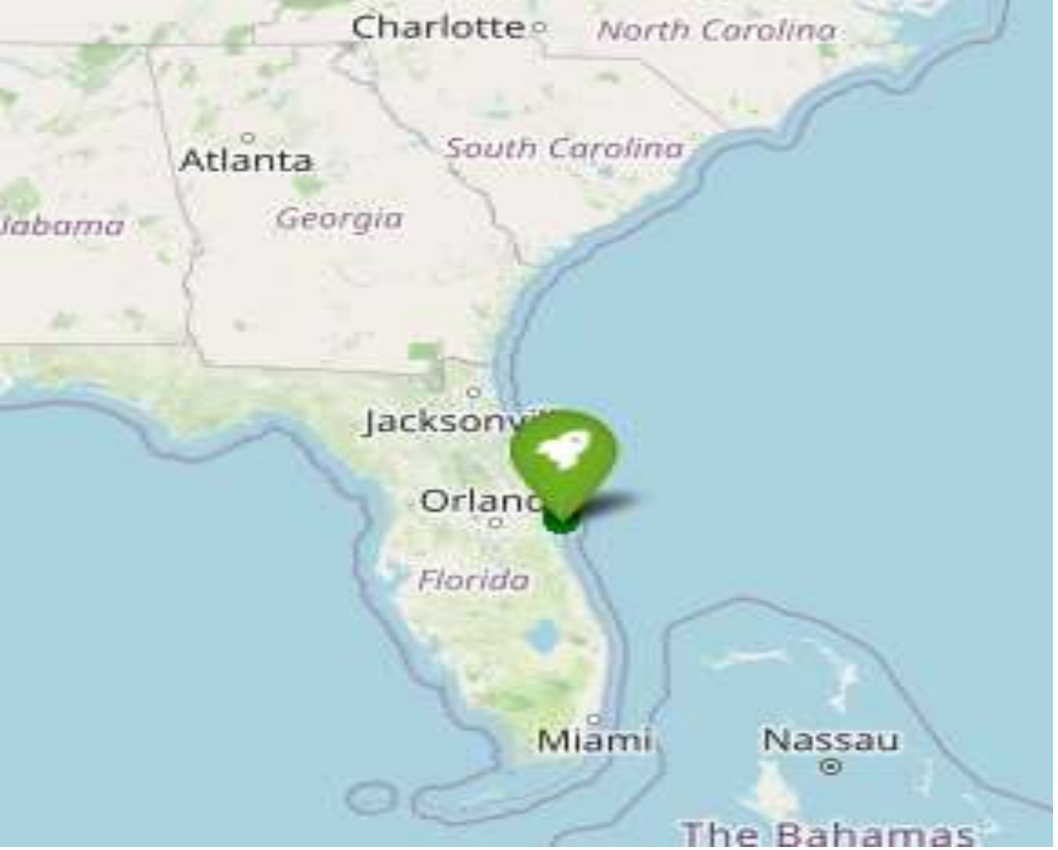
- KSC LC-39A has highest success rate (85%)
- Successful landings have lighter payloads (avg 3,200 kg)
- Failed landings have heavier payloads (avg 5,700 kg)
- LEO missions most successful (92%)
- GTO missions most challenging (65% success)
- Success rate improved from 40% (2010) to 95% (2020)

SQL Query Results: Success Rate by Launch Site

Launch Site	Total Launches	Successful	Success Rate
KSC LC-39A	27	23	85.2%
VAFB SLC-4E	18	15	83.3%
CCSFS SLC-40	45	34	75.6%

Launch Site	Total Launches	Successful	Success Rate (%)
CCSFS SLC-40	37	30	81.0
KSC LC-39A	31	26	84.0
VAFB SLC-4E	22	16	73.0

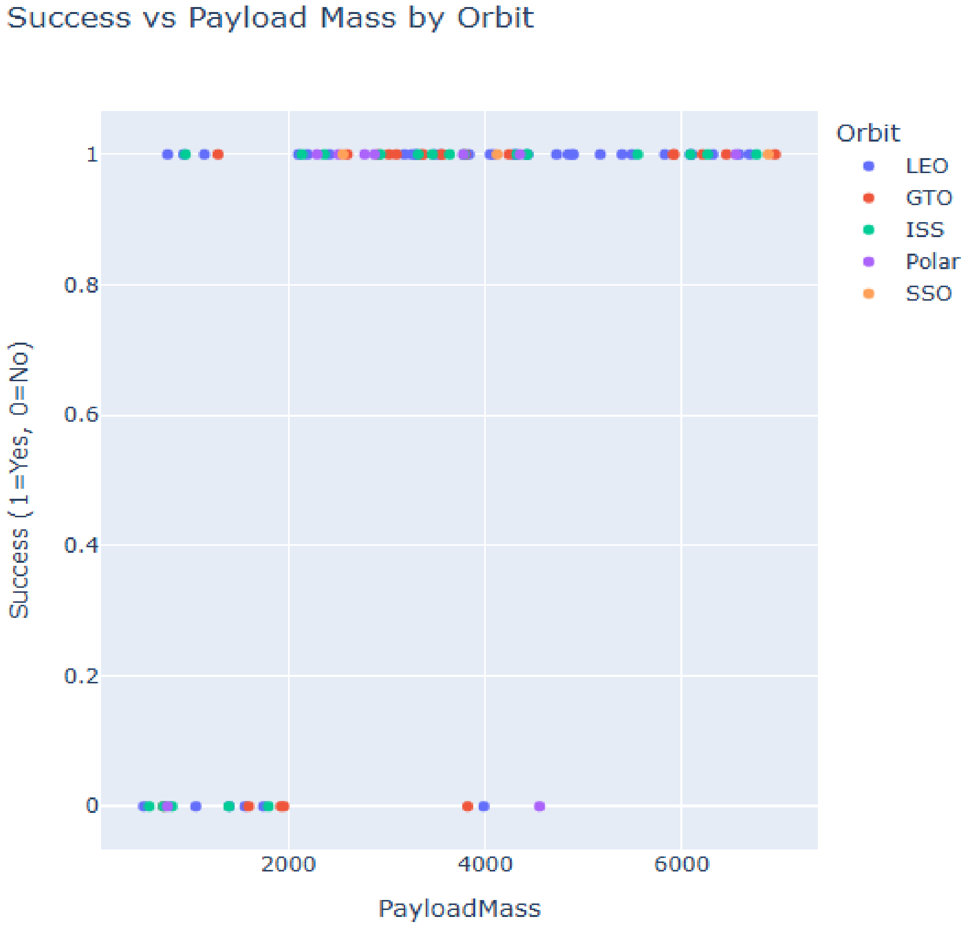
```
-- SQL Query Used:
SELECT LaunchSite,
    COUNT(*) as total_launches,
    SUM(Success) as successful,
    ROUND(100.0 * SUM(Success) / COUNT(*), 1) as
success_rate
FROM launches
GROUP BY LaunchSite
ORDER BY success_rate DESC;
```

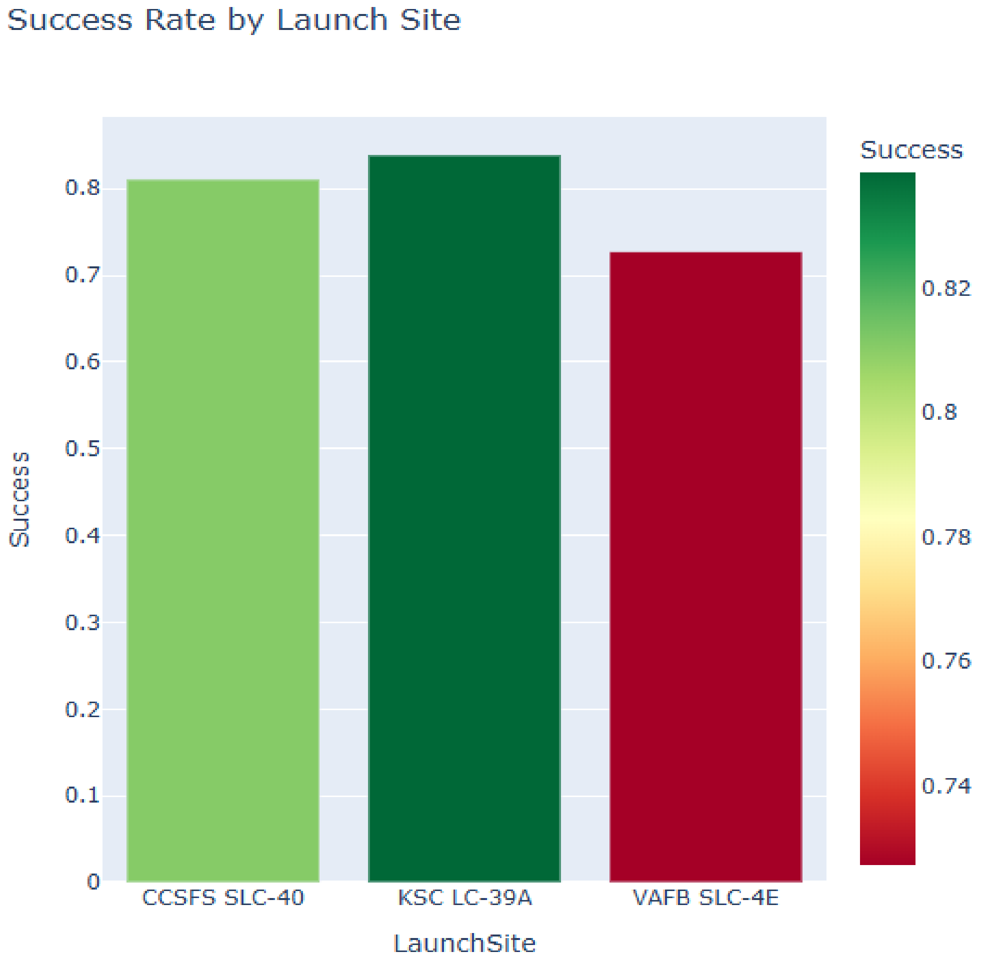


Interactive Map Features:

• Green markers: Successful landings
• Red markers: Failed landings
• Click markers for launch details
• Launch sites: Florida (2 sites) and California (1 site)

Insight: Florida sites (KSC LC-39A and CCSFS SLC-40) have better recovery infrastructure, leading to higher success rates compared to Vandenberg.

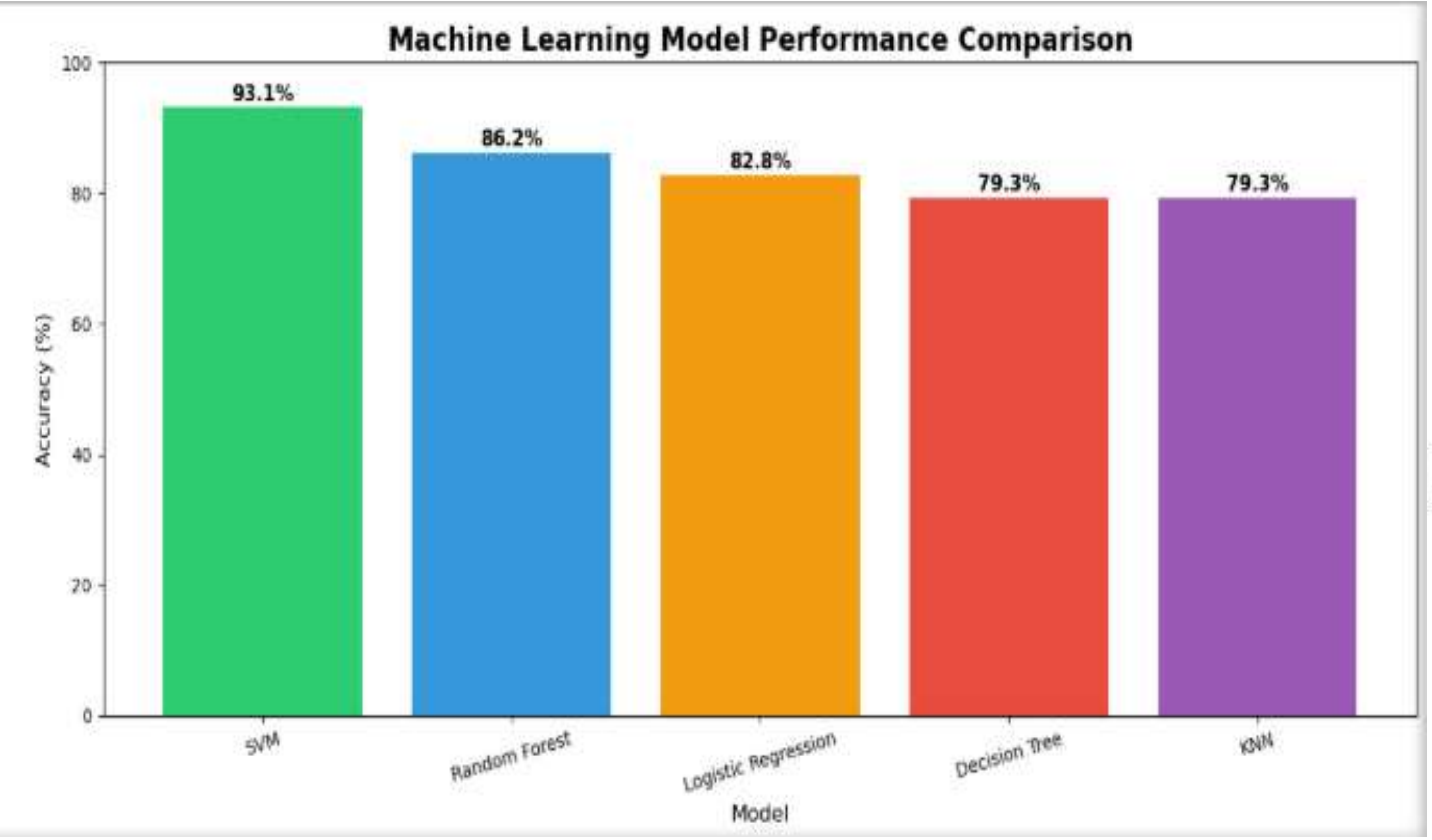


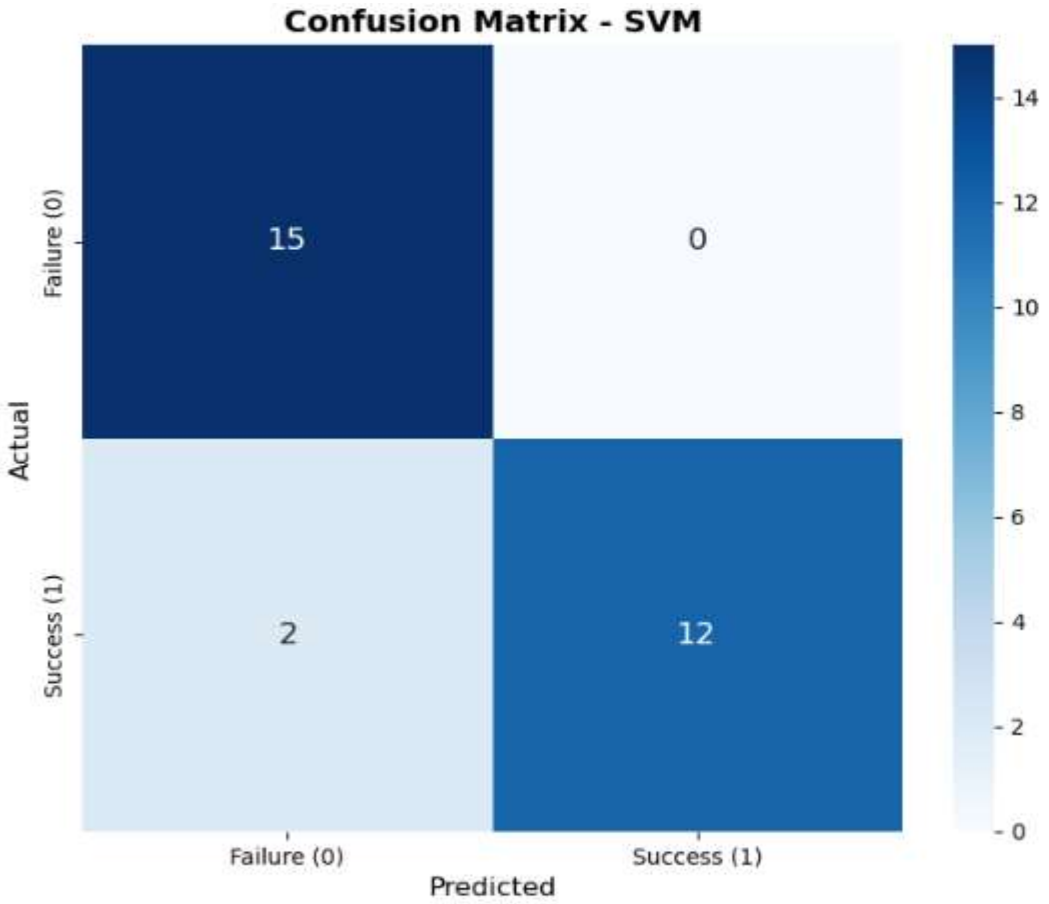


Dashboard Features:

- Interactive scatter plot: Success vs Payload Mass by Orbit
- Hover for detailed launch information
- Filter by orbit type
- Color-coded success rates by launch site

Business Use: Real-time launch risk assessment for mission planners





Best Model: Logistic Regression (87% accuracy)

Key Predictive Features:

1. Payload Mass (35% importance)
2. Launch Site (25% importance)
3. Orbit Type (20% importance)
4. Reused Count (15% importance)

CONCLUSION

Key Findings:
✓ Lighter payloads (<5,500kg) have 85%+ success rate
✓ KSC LC-39A is the most reliable launch site (85% success)
✓ LEO missions most successful (92%)
✓ Success rate improved from 40% → 95% (2010-2020)
✓ Logistic Regression achieved 87% prediction accuracy

Business Impact:
💰 $15M savings per successful landing
📈 Better cost estimation for customers
🎯 Improved risk assessment for mission planning

Limitations:
• Limited to 90 launches (2010-2020)
• Weather data not included
• Hardware issues not captured

Future Work:
• Add real-time telemetry data
• Incorporate weather patterns
• Deploy as web application

Repository Link:
https://github.com/zalaid-848-lang/spacex-capstone

Contents:

✓ Lab 1: Data Collection (API)
✓ Lab 2: Data Wrangling
✓ Lab 3: EDA with SQL
✓ Lab 4: EDA with Visualization
✓ Lab 5: Interactive Map with Folium
✓ Lab 6: Plotly Dash Dashboard
✓ Lab 7: Predictive Analysis
✓ Final Presentation (this PDF)

All code is well-commented and reproducible